eleVADR
Windows Installation and User Guide

From a fresh clone to first analysis

Purpose. This guide walks an operator through installing, starting, signing in to, and using the eleVADR local Windows platform: local MongoDB, account authentication, saved reports, retained PCAPs, and reusable analysis evidence. It covers a fresh clone of the repository as well as a release package.

Currency note. This edition matches the **develop** branch (package version 0.1.0, November 2026). The repository README, the Install page, and the Operator Guide describe the same workflow. The root-level scripts are the authority: if a later release changes them, follow the scripts and release notes shipped with that release. The earlier container-first workflow (`docker compose up --build`) is not used by this platform.

Audience	Windows analysts, evaluators, and local administrators
Platform	Windows 10/11, PowerShell or Command Prompt
Guide scope	Local single-workstation deployment
Validated workflow	Python 3.12.x, Node 22.x, npm 11.x, Docker Desktop
Companion documents	Operator Guide (docs/source/operator-guide.md), Install page (docs/source/install.md), eleVADR_detection_module_reference_updated.pdf, eleVADR_Database_Architecture_and_MongoDB_Schema.pdf

1. What you need before installation

Install the following before running eleVADR for the first time. The first-run preflight checks these dependencies and stops if a required component is missing or incompatible.

Component	Requirement	Why eleVADR needs it
Git	Any recent version (for cloning)	Obtaining the source
Python	3.12 or newer, available on PATH as `python`	Backend, analysis engine, platform services
Node.js	22.22.3 or newer **within the Node 22 line**. Node 23 and later are rejected	Frontend toolchain
npm	Installed with Node	Locked frontend dependency installation
Docker Desktop	Current version, with the engine running	Local MongoDB, and Zeek when no native Zeek is found
Zeek (optional)	8.0.5 or newer on PATH, or set `ELEVADR_ZEEK_COMMAND`	Without it, eleVADR runs the `zeek/zeek:9.0.0` Docker image (override with `ELEVADR_ZEEK_DOCKER_IMAGE`)
Disk space	Several GB free, plus space for retained PCAPs	Dependencies, MongoDB, capture/report storage

Verify the tools

Open PowerShell and run:

```
git --version
python --version
node --version
npm --version
docker --version
docker info
```

Docker Desktop must be running. A successful `docker info` confirms the engine is available.

2. Get eleVADR

Option A: clone the repository

Clone into a short path that is **not** inside a cloud-synced folder such as OneDrive. Syncing `node_modules` can slow installs and leave partially written dependencies. `develop` is the default branch.

```
git clone https://github.com/cisagov/eleVADR.git C:\dev\eleVADR
cd C:\dev\eleVADR
```

Option B: release package

1. Obtain the current eleVADR release/package ZIP.
2. Extract it to a normal writable folder, for example `C:\eleVADR`. Do not run the application from inside the ZIP.
3. Open PowerShell in the extracted root, the directory that contains the root-level `.bat` files.

If you use a Python virtual environment, create and activate it in this shell first. The launcher windows inherit the shell environment, so use the same shell for the scripts that follow.

3. First-run setup

Run the first-run script once from the eleVADR root:

```
.\first_run_elevadr.bat
```

The script stops at the first failure, so its last message tells you which step to fix. It performs these steps:

1. Runs the deployment preflight (Python, Node, npm, Docker, and required package files).
2. Installs the platform Python requirements from `backend\requirements-platform.txt` with pip.
3. Installs the frontend from `frontend\package-lock.json` with `npm ci`.
4. Prepares the local database: generates `.elevadr-platform.env` with random MongoDB credentials and a JWT signing secret, starts the localhost-only MongoDB container, and waits until it is healthy.
5. Prompts for the first administrator if no users exist (default username `admin`). Passwords must be at least 12 characters.
6. Enables authentication and reruns the preflight.

A successful run ends with **PASS: eleVADR first-run setup completed** and tells you to start the application with `start_elevadr.bat`. The script is safe to run again on an initialized installation: existing users are detected and the administrator prompt is skipped.

Optional: pre-pull the Zeek image

If you do not have a native Zeek installation, pull the Docker image now so the first PCAP analysis does not depend on network access:

```
.\prepare_zeek_runtime.bat
```

Protect the generated configuration

`.elevadr-platform.env` holds the MongoDB credentials and JWT signing secret. It is excluded from git. Never commit it, copy it between machines, or paste it into support logs. A documented template without secrets is provided as `.elevadr-platform.env.example`.

4. Start, stop, and daily operation

Start

```
.\start_elevadr.bat
```

The launcher opens two command windows, one for the backend and one for the frontend. Keep both open while using eleVADR, and check them for startup errors. The frontend runs at `http://127.0.0.1:5173/` and the backend at `http://127.0.0.1:8765/`.

The script also acts as a restart. It stops any eleVADR frontend or backend already listening on ports 5173 and 8765, removes leftover eleVADR Zeek containers, and aborts if a different program is using either port. In-memory processing jobs are lost on restart; MongoDB data, saved reports, retained PCAPs, and contexts are kept. It warns if the platform environment file is missing.

MongoDB is not started by this script. If the database is not running, for example after Docker Desktop was stopped, run `start_elevadr_database.bat` first.

Check the backend

```
Invoke-RestMethod http://127.0.0.1:8765/health
```

The health endpoint reports detector, contract, and Zeek-runtime status.

Stop

```
.\stop_elevadr.bat
```

This closes the eleVADR frontend and backend windows and ends any process still listening on port 8765. To stop the database as well, run `stop_elevadr_database.bat`. The data is preserved.

Do not run `docker compose down -v`. It deletes the MongoDB volume and all users, report metadata, and audit history.

Task	Command
Start or restart backend and frontend	`.\start_elevadr.bat`
Stop backend and frontend	`.\stop_elevadr.bat`
Start the database	`.\start_elevadr_database.bat`
Stop the database (data preserved)	`.\stop_elevadr_database.bat`
Check database health	`.\check_elevadr_database.bat`

5. What a fresh clone does not contain

The repository holds source code only. These items are created on each machine and are intentionally excluded from git.

Item	How it is created	Notes
`.elevadr-platform.env`	First-run database setup	Secrets. Keep it with its database (see below).
`frontend\node_modules`	`npm ci` during first run	Rerun `npm ci` in `frontend` if missing or damaged.
Python packages	pip during first run	Installed into the active Python environment.
MongoDB data	Docker volume `elevadr-mongodb-data`	Users, report index, and audit history.
`storage\`	Created by the backend	Retained PCAPs and report JSON under `storage\users\`.
Zeek image	On first use, or `prepare_zeek_runtime.bat`	Not needed when native Zeek is on PATH.

`frontend\.env.local` ships with the repository and points the frontend at the local backend on port 8765. Edit it only if the backend port changes, then restart the frontend.

Keep the environment file and the database together

MongoDB applies its root credentials only when the data volume is first created. If `.elevadr-platform.env` is deleted and regenerated while the `elevadr-mongodb-data` volume still exists, the new credentials will not match the database and the backend cannot authenticate. Keep the file, or restore from a backup into a fresh setup.

Moving to a new clone or machine

A new clone starts with an empty database and empty storage. To carry over users, reports, and retained PCAPs, use backup and restore (section 12).

6. Sign in and understand the workspace

Open `http://127.0.0.1:5173/` in a browser. When authentication is enabled, the welcome screen presents the sign-in form. Enter the administrator or analyst credentials created for you.

After sign-in, the navigation provides Dashboard, Reports, PCAPs, Audit/Activity, Storage, and Account & Users/Account. Labels and administrative functions depend on the account role. The backend enforces roles; hiding a control in the interface is not the security boundary.

Capability	Admin	Analyst	Read Only
View/open own saved reports	Yes	Yes	Yes
View retained PCAP metadata	Yes	Yes	Yes
Upload/analyze PCAP; re-analyze retained PCAP	Yes	Yes	No
Change Analysis Context/Modules	Yes	Yes	No
Rename/delete own reports; delete retained PCAP	Yes	Yes	No
Change own password	Yes	Yes	Yes
Manage users and roles	Yes	No	No

7. Analyze a PCAP

1. Sign in and return to the Dashboard.
2. Use the upload area to select or drag in a `.pcap` or `.pcapng` file. The default upload limit is 256 MiB.
3. Review the Detection Context before analysis. Context represents site-specific policy and expectations such as network scope, assets, expected communications, and OT/Purdue information.
4. Review Modules if you need to enable or disable detection modules. The standard registry contains 75 detector modules.
5. Start the analysis. eleVADR runs Zeek evidence extraction and the selected detection modules, then builds the report.
6. When complete, review the report: Summary, Findings, Devices, Services, and Connections. Use the left navigation and the search/pivot controls to investigate.

Evidence reuse. Zeek evidence from context discovery is reused for the analysis path rather than deliberately rerun, and retained evidence can support re-analysis with a different Detection Context. If a PCAP is rejected as too large, an operator can adjust `ELEVADR_MAX_PCAP_UPLOAD_MB` deliberately rather than removing the bound.

8. Work with reports

Saved reports

Completed PCAP analyses are saved automatically for the signed-in user as immutable reports. Open Reports to search and sort them. Select the report name or details area to open one. The adjacent icons provide rename and delete when your role permits. Report JSON is stored under `storage\users\<user-id>\reports\`.

Inside a report

- Use the left navigation to move between Summary, Findings, Devices, Services, and Connections.
- Use search and pivots to filter by entities such as IP, service, port, subnet, manufacturer, or risk.
- Use Customize to control visible report sections.
- Use Print or Export for output. Export can create a filtered derivative view; the source report is not modified.
- Use Modules or Context to change analysis configuration. If the configuration changes while a report is open, eleVADR prompts you to generate an updated report.

9. Retained PCAPs, history, and comparison

Open PCAPs to see captures retained for your account, stored under `storage\users\<user-id>\pcaps\`. Identical captures are deduplicated per user by SHA-256. Each row shows capture metadata and the number of saved analyses.

- History icon: show or hide prior analyses for the capture.
- Re-analyze icon: run the analysis again with the active Detection Context and module selection, without uploading the PCAP again.
- Delete icon: remove the retained capture, subject to permissions and confirmation. Deleting or expiring a PCAP never deletes its saved reports.
- Select two analyses from the same capture to compare new, resolved, and changed findings, severity/confidence changes, assets, services, connections, Context and module changes, and runtime metadata.

10. Detection Context and Modules

Detection Context

Context is where eleVADR learns what is expected for the capture or site. Complete as much of the wizard as is known. Marking normal behavior as unknown can increase findings; overly broad allowlisting can suppress useful detections. Saved Context profiles are owner-scoped, and applying or deleting a profile does not change the Context snapshot already embedded in a completed report.

Detection Modules

Modules control which detector families execute. Leave the standard selection intact unless you have a reason to tailor analysis. When reviewing a finding, use its provenance information to see which Zeek log type and supporting fields contributed to the result.

11. Account, users, storage, and audit

Account and users

Use Account to change your password. New passwords must be at least 12 characters, and changing a password ends existing sessions for that account. Administrators also see Account & Users, where they can create users, assign roles, and enable or disable accounts. An administrator cannot disable their own signed-in account. Login failures are throttled; there is no permanent automatic lockout.

Storage and retention

The Storage view summarizes retained capture storage, report storage, retention policy, and limits. Administrators can review per-user usage and run cleanup for expired captures and, when requested, orphaned capture files. Two settings in `.elevadr-platform.env` control retention; a value of 0 disables each one:

```
ELEVADR_PCAP_RETENTION_DAYS=0
ELEVADR_PCAP_STORAGE_LIMIT_GB=0
```

Audit / Activity

Administrators can review platform-wide audit activity. Other users see activity for their own account. Audit records use bounded, sanitized metadata, and password-, token-, secret-, and raw-PCAP-like fields are redacted.

12. Backup and restore

Backups include the MongoDB platform documents and `storage\users\` so metadata and retained files move together. JWT and MongoDB secrets, browser sessions, Docker internals, and temporary analysis files are excluded, so a restored installation keeps its own secrets and users sign in again.

```
.\backup_elevadr.bat
```

By default archives are not encrypted; protect them as sensitive data. To encrypt, set `ELEVADR_BACKUP_PASSPHRASE` in the environment used for backup. The same passphrase is required to validate and restore. Validate a backup without changing local state:

```
.\validate_elevadr_backup.bat <backup-file>
```

Before a restore, stop the backend (`stop_elevadr.bat`) but leave MongoDB running. Restore is destructive and requires explicit acknowledgement:

```
.\restore_elevadr.bat <backup-file> --yes
```

After a restore, restart eleVADR and run the platform regression gate. To move to a new clone: back up on the old installation, complete first-run setup on the new one, then restore.

13. Validate the installation

On an initialized workstation, run:

```
.\validate_clean_install.bat
```

It runs the initialized preflight, the database health check, a frontend build, the release preflight, and the platform regression tests, so it takes noticeably longer than other steps. Two related scripts are available from the repository root:

- `run_platform_regression_tests.bat`: the quick platform and security check.
- `run_regression_tests.bat`: the complete detector, PCAP, UI, compatibility, performance, and platform gate (maintainer-oriented).

Treat the PASS or FAIL result, not a hard-coded test count, as authoritative.

14. Troubleshooting

Symptom	What to check
Preflight rejects Node.js	Install a supported 22.x release (22.22.3 or newer, below 23).
Docker or MongoDB setup fails	Start Docker Desktop, confirm `docker info` succeeds, then run `check_elevadr_database.bat` and `start_elevadr_database.bat` or retry first-run.
Backend cannot connect to MongoDB or cannot authenticate	Confirm the database is running. If `.elevadr-platform.env` was regenerated while the old volume exists, restore the original file or rebuild from a backup.
Frontend does not start, or reports a missing `vite` module	`node_modules` is missing or incomplete. In `frontend` run `npm ci` (or rerun first-run). Use npm for the local platform; do not mix installers in one `node_modules` folder. Avoid OneDrive-synced folders.
Frontend dependencies fail	Use the package lock and supported Node/npm versions. Avoid `--force` or `--legacy-peer-deps` unless intentionally documented.
Port 5173 or 8765 occupied	The start script stops stale eleVADR listeners but aborts if another program holds the port. Close that program and retry.
Warning that the platform environment file is missing	Run `setup_elevadr_database.bat` (or first-run) to generate it.
PCAP analysis cannot start	Verify Docker is running, the Zeek image is available, or native Zeek is on PATH. Viewing a JSON report works without Zeek.
Browser shows *Failed to fetch*	Confirm the backend is on 8765 and the frontend on 5173. Custom deployments must set `ELEVADR_CORS_ORIGINS`.
PCAP rejected as too large	The default limit is 256 MiB. Adjust `ELEVADR_MAX_PCAP_UPLOAD_MB` deliberately.
Cannot sign in	Confirm first-run created a user, authentication is enabled, credentials are correct, and the backend window shows no error.
Backup validation fails	Do not restore the archive. Check the passphrase, create a new backup, or investigate the failed archive.
Python package conflicts	Use a dedicated virtual environment. In a shared Python installation, unrelated packages can cause `pip check` conflicts.
UI looks stale after replacing files	Restart the frontend and refresh the browser without cache.

15. Clean-install quick checklist

- **1.** Install Git, Python 3.12+, Node 22.22.3+ (22.x), and Docker Desktop.
- **2.** Start Docker Desktop and verify `docker info`.
- **3.** Clone the repository outside OneDrive (or extract the release package).
- **4.** Open PowerShell in the eleVADR root.
- **5.** Run `.\first_run_elevadr.bat`.
- **6.** Create the first administrator when prompted (new install only).
- **7.** Confirm first-run reports PASS.
- **8.** Optional: run `.\prepare_zeek_runtime.bat` if you use the Docker Zeek runtime.
- **9.** Run `.\start_elevadr.bat`.
- **10.** Open `http://127.0.0.1:5173/` and sign in.
- **11.** Review Context and Modules, then analyze a PCAP.
- **12.** Confirm the report is saved and the PCAP appears under PCAPs.
- **13.** When finished, run `.\stop_elevadr.bat` (and `.\stop_elevadr_database.bat` if desired).
- **14.** After a reboot or Docker restart, run `.\start_elevadr_database.bat` before `.\start_elevadr.bat`.

16. Source and version notes

This guide follows the local Windows platform workflow in the **develop** branch. It is consistent with the repository README, `docs/source/install.md`, and `docs/source/operator-guide.md`. If a future release changes the root-level scripts, follow the scripts and release notes shipped with that release.

Companion references in the repository: `eleVADR_detection_module_reference_updated.pdf` for detector-specific behavior, and `eleVADR_Database_Architecture_and_MongoDB_Schema.pdf` for the database design.